

DAY – 8

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Python [conda env:base] Anaconda Toolbox

OUTLIERS

```
[26]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import zscore

#Load titanic dataset
df=sns.load_dataset('titanic')

#box plot
sns.boxplot(x=df['fare'])
plt.show()

#IQR Method
Q1=df['fare'].quantile(0.25)
Q3=df['fare'].quantile(0.75)
IQR=Q3-Q1
lower_limit=Q1 - 1.5 * IQR
upper_limit=Q3 + 1.5 * IQR

outliers_iqr=df[
    (df['fare'] < lower_limit) |
    (df['fare'] > upper_limit)]
print("IQR Outliers:",len(outliers_iqr))
```

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IQR Outliers: 116

```
[30]: ## Z-Score Method
df['fare_zscore']=zscore(df['fare'])
outliers_z=df[
    abs(df['fare_zscore'])>3
]
print("Z-score Outliers:",len(outliers_z))
Z-score Outliers: 20
```

EDA

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Python [conda env:base] Anaconda Toolbox

EDA

```
[31]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
print(df.head())
print(df.info())
print(df.columns)
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
Column Non-Null Count Dtype

0 total_bill 244 non-null float64
1 tip 244 non-null float64
2 sex 244 non-null category
3 smoker 244 non-null category
4 day 244 non-null category
5 time 244 non-null category
6 size 244 non-null int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
None
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')

```
[32]: #Descriptive statistics
print(df['total_bill'].describe())
```









